Lecture 15

Strong Continuity and Infinitesimal Generators

Objectives. *Part II studies unitary representations on infinite-dimensional Hilbert spaces. Their derivatives may exist only on a proper subspace. This lecture fixes the continuity hypothesis, defines the derivative on smooth vectors, and uses Stone's theorem to identify the skew-adjoint generator of each one-parameter subgroup, together with its domain. Translations on $L^2(\mathbb{R})$ are the model throughout.*

By the end of this lecture you will be able to:

- Prove strong continuity of a homomorphism into the unitary group from continuity at the identity on a dense set.
- Define smooth vectors and $d\Pi$, and prove that each $d\Pi(X)$ is skew-symmetric on smooth vectors.
- State Stone's theorem with its hypotheses, and identify the closure of $d\Pi(X)$ with the generator.
- Locate explicit vectors relative to the domains for translations: Schwartz functions, a non-Schwartz vector in the domain, and a vector outside it.

15.1 From matrices to Hilbert spaces

Let $\Pi\colon G \to \mathrm{GL}(V)$ be a continuous representation on a finite-dimensional complex space V. Continuous homomorphisms of Lie groups are smooth (Lecture 3). So every $X \in \mathfrak{g}$ gives an operator

$$d\Pi(X) = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX) \in \mathrm{End}(V), \qquad \Pi(\exp tX) = e^{t\,d\Pi(X)}.$$

If Π is unitary, then $d\Pi(X)^* = -d\Pi(X)$. Every vector can be differentiated.

For compact G, the Peter–Weyl theorem (Lecture 10) decomposes each unitary representation into finite-dimensional subrepresentations. Noncompact groups behave differently. The representations studied in Part II are infinite-dimensional, and the derivative may be defined only on a proper subspace. The basic example is the translation representation on $L^2(\mathbb{R})$, so we first construct that space.

Let $C_c(\mathbb{R})$ be the space of continuous functions $\mathbb{R} \to \mathbb{C}$ with compact support, with inner product

$$\langle f, g\rangle = \int_{\mathbb{R}} \overline{f(x)}\, g(x)\, dx.$$

Its Hilbert completion is $L^2(\mathbb{R})$.

We use one fact about $L^2(\mathbb{R})$. A *jump* of a function $f\colon \mathbb{R} \to \mathbb{C}$ is a point where f is discontinuous but has both one-sided limits. We call f *piecewise continuous* if it is continuous except at finitely many jumps in each bounded interval. If moreover the improper integral $\int_{\mathbb{R}} |f|^2\,dx$ is finite, we call f *square integrable*. Such a function f defines a vector of $L^2(\mathbb{R})$, and inner products of these vectors are given by the same integral. We first treat functions that vanish outside a bounded interval. Then we pass to square integrable functions by truncation.

Let f be piecewise continuous and vanish outside a bounded interval. Linear interpolation across the jumps of f gives functions $g_n \in C_c(\mathbb{R})$ with $\int_{\mathbb{R}} |f - g_n|^2\,dx \to 0$. The sequence (g_n) is Cauchy in $L^2(\mathbb{R})$. We call its limit the vector f. This vector does not depend on the approximating sequence, and it depends linearly on f. For two such functions f and $\tilde f$ we have $\|f\|^2 = \int_{\mathbb{R}} |f|^2\,dx$ and $\langle f, \tilde f\rangle = \int_{\mathbb{R}} \bar f \tilde f\,dx$. The preliminary note in the solution of Exercise 15.1 proves these statements. It constructs the functions g_n explicitly and uses the Cauchy–Schwarz inequality for Riemann integrals.

Now let f be square integrable. Each truncation $f \cdot 1_{[-n,n]}$ is piecewise continuous and vanishes outside $[-n, n]$, so it defines a vector. The difference of two truncations is $f \cdot 1_{\{n<|x|\le m\}}$. By linearity and the norm formula, the truncations form a Cauchy sequence, because

$$\left\| f \cdot 1_{[-m,m]} - f \cdot 1_{[-n,n]} \right\|^2 = \int_{n<|x|\le m} |f|^2\,dx \longrightarrow 0 \qquad (m > n \to \infty).$$

Their limit is the vector f. For square integrable f and $\tilde f$, letting $n \to \infty$ in the formulas for the truncations gives

$$\|f\|^2 = \int_{\mathbb{R}} |f|^2\,dx, \qquad \langle f, \tilde f\rangle = \int_{\mathbb{R}} \bar f \tilde f\,dx.$$

The second improper integral converges absolutely, since $\int_{-n}^{n} |\bar f \tilde f|\,dx \le \|f\|\,\|\tilde f\|$ by the Cauchy–Schwarz inequality.

Changing f at finitely many points does not change the vector. Indeed, the difference of the two functions is nonzero at only finitely many points, so its vector has norm 0.

For $t \in \mathbb{R}$ and $f \in C_c(\mathbb{R})$ put

$$(\lambda(t) f)(x) = f(x - t).$$

Then $\lambda(s)\lambda(t) = \lambda(s+t)$, and $\|\lambda(t) f\| = \|f\|$ by substitution. So each $\lambda(t)$ extends to a unitary operator on $L^2(\mathbb{R})$ with inverse $\lambda(-t)$. On the piecewise continuous functions above, $\lambda(t)$ is still translation.

We identify the Lie algebra of $\mathbb{R}$ with $\mathbb{R}$. With this identification the exponential map $\mathbb{R} \to \mathbb{R}$ is the identity, so $d\lambda(1) f$ is the derivative of $t \mapsto \lambda(t) f$ at $t = 0$. Formally $\frac{d}{dt}\lambda(t) f\big|_{t=0} = -f'$. This formula has no meaning for a general vector. In the domain example, Example 15.15, the difference quotients of $1_{[0,1]}$ are unbounded. So $d\lambda(1)$ needs a stated domain.

15.2 Strong continuity

Definition 15.1. Let $\mathcal{H}$ be a complex Hilbert space and $\mathrm{U}(\mathcal{H})$ its group of unitary operators. A *unitary representation* of a Lie group G on $\mathcal{H}$ is a homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ such that the map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is continuous for every $v \in \mathcal{H}$. This condition is *strong continuity*. The homomorphism is *norm continuous* if $g \mapsto \Pi(g)$ is continuous for the operator norm.

Lemma 15.2. *Let $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ be a homomorphism. Let $D \subset \mathcal{H}$ be dense, and suppose $g \mapsto \Pi(g)w$ is continuous at e for every $w \in D$. Then Π is strongly continuous.*

Proof. Let $v \in \mathcal{H}$ and $w \in D$. Since $\Pi(g)$ is unitary,

$$\|\Pi(g)v - v\| \le \|\Pi(g)(v-w)\| + \|\Pi(g)w - w\| + \|w - v\| = 2\|v-w\| + \|\Pi(g)w - w\|.$$

Given $\varepsilon > 0$, choose w with $\|v - w\| < \varepsilon$. Then $\limsup_{g\to e} \|\Pi(g)v - v\| \le 2\varepsilon$. So $g \mapsto \Pi(g)v$ is continuous at e. At another point h we use

$$\|\Pi(g)v - \Pi(h)v\| = \|\Pi(h^{-1}g)v - v\|,$$

and $h^{-1}g \to e$ as $g \to h$. ∎

Proposition 15.3. *The translations λ form a unitary representation of $\mathbb{R}$ on $L^2(\mathbb{R})$.*

Proof. Only strong continuity remains. We apply the dense-set criterion, Lemma 15.2, with $D = C_c(\mathbb{R})$. Let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$. For $|t| \le 1$ the function $\lambda(t)f - f$ vanishes outside $[-R-1, R+1]$. Hence

$$\|\lambda(t)f - f\|^2 \le (2R+2) \sup_{x\in\mathbb{R}} |f(x-t) - f(x)|^2.$$

Since f is continuous with compact support, it is uniformly continuous. Hence the right side tends to 0 as $t \to 0$, and the dense-set criterion gives strong continuity. ∎

Norm continuity would be too strong a requirement, since translations are not norm continuous. Indeed, Exercise 15.1 shows that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Strong continuity requires continuity separately for each vector. It does not require this continuity to be uniform over unit vectors.

15.3 Smooth vectors and the derived representation

A map from an open subset of $\mathbb{R}^n$ to $\mathcal{H}$ is *smooth* if all its partial derivatives of all orders exist as limits in the norm of $\mathcal{H}$ and are continuous. A map $G \to \mathcal{H}$ is smooth if it is smooth in every chart.

Definition 15.4. Let Π be a unitary representation of G on $\mathcal{H}$. A vector $v \in \mathcal{H}$ is *smooth* if $g \mapsto \Pi(g)v$ is a smooth map $G \to \mathcal{H}$. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ put

$$d\Pi(X)v = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX)v = \lim_{t \to 0} \frac{\Pi(\exp tX)v - v}{t}.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, since linear combinations of smooth maps $G \to \mathcal{H}$ are smooth. The limit defining $d\Pi(X)v$ exists because $t \mapsto \Pi(\exp tX)v$ is the smooth curve $t \mapsto \exp tX$ followed by the smooth map in the definition of a smooth vector.

Proposition 15.5. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$, and let $g \in G$, $X \in \mathfrak{g}$.*

(a) $\Pi(g)\mathcal{H}^\infty \subset \mathcal{H}^\infty$ *and* $d\Pi(X)\mathcal{H}^\infty \subset \mathcal{H}^\infty$.

(b) $\mathcal{H}^\infty$ *is dense in* $\mathcal{H}$.

(c) $X \mapsto d\Pi(X)$ *is a representation of* $\mathfrak{g}$ *on* $\mathcal{H}^\infty$, *and* $d\Pi(\mathrm{Ad}(g)X) = \Pi(g)\, d\Pi(X)\, \Pi(g)^{-1}$ *on* $\mathcal{H}^\infty$.

(d) For $v, w \in \mathcal{H}^\infty$ *we have* $\langle d\Pi(X)v, w\rangle = -\langle v, d\Pi(X)w\rangle$.

Proof. We cite Knapp, *Representation Theory of Semisimple Groups*, Sections III.3–4, for the invariance under $d\Pi(X)$ in (a), the density in (b), and the representation property and Ad formula in (c). We prove the rest.

For the invariance under $\Pi(g)$ in (a), let $v \in \mathcal{H}^\infty$, $h \in G$ and $F(g) = \Pi(g)v$. Then $\Pi(g)\Pi(h)v = F(gh)$. Right translation by h is smooth on G, so $g \mapsto F(gh)$ is smooth.

For the skew-symmetry in (d), unitarity gives

$$\langle \Pi(\exp tX)v, w\rangle = \langle v, \Pi(\exp(-tX))w\rangle.$$

Subtract $\langle v, w\rangle$ from both sides and divide by $t \neq 0$:

$$\left\langle \frac{\Pi(\exp tX)v - v}{t},\, w \right\rangle = -\left\langle v,\, \frac{\Pi(\exp(-tX))w - w}{-t} \right\rangle.$$

The inner product is continuous, so letting $t \to 0$ gives (d). ∎

The omitted density step in (b) is the hard one, so we sketch it. The sketch uses a vector-valued integral against a left Haar measure dg on G. For compact G, Haar measure is introduced in Lecture 5. For noncompact G it is introduced in Lecture 16, so until then we cite Knapp for this integral. For $\varphi \in C_c^\infty(G)$ and $v \in \mathcal{H}$ form the *Gårding vector*

$$\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(g)v \, dg.$$

Let $h \in G$. Since $\Pi(h)$ is bounded, it passes inside the integral. Left invariance of dg then allows the substitution $g \mapsto h^{-1}g$. It gives

$$\Pi(h)\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(hg)v \, dg = \int_G \varphi(h^{-1}g)\, \Pi(g)v \, dg.$$

Now h appears only inside the smooth scalar function $(h,g)\mapsto \varphi(h^{-1}g)$. Differentiating under the integral in local coordinates of h shows that $h\mapsto \Pi(h)\Pi(\varphi)v$ is smooth on G. So $\Pi(\varphi)v$ is a smooth vector. If $\varphi_n\ge 0$, $\int_G \varphi_n\,dg=1$ and the supports shrink to e, then

$$\|\Pi(\varphi_n)v-v\|\le \sup_{g\in \operatorname{supp}\varphi_n}\|\Pi(g)v-v\|\longrightarrow 0$$

by strong continuity. Exercise 15.4 carries out the argument for $G=\mathbb{R}$ with Riemann integrals.

15.4 Unbounded operators and Stone's theorem

On $L^2(\mathbb{R})$ the operator $-d/dx$ is defined only on a subspace. We need language for such operators.

Definition 15.6. An *operator* A on $\mathcal{H}$ is a linear map $A\colon D(A)\to\mathcal{H}$ defined on a subspace $D(A)\subset\mathcal{H}$, its *domain*. It is *densely defined* if $D(A)$ is dense. We write $B\subset A$ if $D(B)\subset D(A)$ and $Bv=Av$ for $v\in D(B)$.

Definition 15.7. Let A be densely defined. The domain of the *adjoint* A^* is the set of $w\in\mathcal{H}$ for which some $u\in\mathcal{H}$ satisfies

$$\langle w, Av\rangle=\langle u,v\rangle\qquad\text{for all } v\in D(A),$$

and then $A^*w=u$. The operator A is *skew-symmetric* if $\langle Av,w\rangle=-\langle v,Aw\rangle$ for all $v,w\in D(A)$. It is *skew-adjoint* if $A^*=-A$, including $D(A^*)=D(A)$, and *self-adjoint* if $A^*=A$, including $D(A^*)=D(A)$.

Since $D(A)$ is dense, the vector u is unique. Skew-symmetry of A means exactly that $-A\subset A^*$.

Suppose $B\subset A$ and B is densely defined. Then $D(A)\supset D(B)$ is dense too, so both adjoints exist. In this case $A^*\subset B^*$. Indeed, the defining identity for A^*w holds in particular for $v\in D(B)$.

Lemma 15.8. *Let B be densely defined and skew-symmetric. If $v_n\in D(B)$, $v_n\to 0$ and $Bv_n\to u$, then $u=0$.*

Proof. For every $w\in D(B)$, continuity of the inner product and skew-symmetry of B give

$$\langle w,u\rangle=\lim_n\langle w,Bv_n\rangle=-\lim_n\langle Bw,v_n\rangle=0.$$

So u is orthogonal to the dense subspace $D(B)$, and $u=0$. ∎

Definition 15.9. Let B be densely defined and skew-symmetric. The domain of its *closure* $\bar B$ is the set of $v\in\mathcal{H}$ for which there are $v_n\in D(B)$ with $v_n\to v$ and (Bv_n) convergent. Then $\bar Bv=\lim_n Bv_n$.

The value $\bar Bv$ does not depend on the sequence. Indeed, the difference of two such sequences satisfies the hypotheses of the limit lemma for skew-symmetric operators, Lemma 15.8.

Definition 15.10. Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a unitary representation. Its *generator* A has domain

$$D(A) = \left\{ v \in \mathcal{H} : \lim_{t \to 0} \frac{U(t)v - v}{t} \text{ exists in } \mathcal{H} \right\},$$

and Av is this limit.

Theorem 15.11 (Stone). *Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a strongly continuous unitary representation with generator A.*

(a) A is densely defined and skew-adjoint. Moreover $U(t)D(A) \subset D(A)$ and $AU(t)v = U(t)Av$ for $v \in D(A)$.

(b) Let $D \subset D(A)$ be a dense subspace with $U(t)D \subset D$ for all t. Then $A|_D$ is skew-symmetric, and its closure is A.

(c) Every skew-adjoint operator is the generator of exactly one strongly continuous unitary representation of $\mathbb{R}$, written $t \mapsto e^{tA}$.

We cite most of this theorem and prove only the invariance statement in (a). For $v \in D(A)$, since $U(t)$ is bounded and commutes with $U(h)$,

$$\frac{U(h)U(t)v - U(t)v}{h} = U(t)\,\frac{U(h)v - v}{h} \longrightarrow U(t)Av \qquad (h \to 0).$$

So $U(t)v \in D(A)$ and $AU(t)v = U(t)Av$.

Hall, *Quantum Theory for Mathematicians*, Section 10.2, Theorem 10.15, gives Stone's theorem. Hall writes $U(t) = e^{itH_{\mathrm{Hall}}}$ with a self-adjoint operator H_{Hall}, so $H_{\mathrm{Hall}} = -iA$. Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4, Theorems VIII.10–11, gives the dense invariant-domain criterion in (b).

Our physical convention writes $U(t) = e^{-itH}$ with a self-adjoint operator H, so $H = iA$ and $H_{\mathrm{Hall}} = -H$.

Corollary 15.12. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$. Let $X \in \mathfrak{g}$, and let B_X be the generator of $t \mapsto \Pi(\exp tX)$. Then $\mathcal{H}^\infty \subset D(B_X)$ and $B_X = d\Pi(X)$ on $\mathcal{H}^\infty$. The closure of $d\Pi(X)$ with domain $\mathcal{H}^\infty$ is the skew-adjoint operator B_X, and $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$.*

Proof. The map $t \mapsto \exp tX$ is a continuous homomorphism $\mathbb{R} \to G$. Composing it with Π gives a homomorphism $\mathbb{R} \to \mathrm{U}(\mathcal{H})$. For each v, the map $t \mapsto \Pi(\exp tX)v$ is a continuous map followed by the continuous map $g \mapsto \Pi(g)v$. So $t \mapsto \Pi(\exp tX)$ is a strongly continuous unitary representation of $\mathbb{R}$, and its generator B_X is defined.

The first two claims restate the definition of $d\Pi(X)$. By the invariance and density statements of Proposition 15.5, $\mathcal{H}^\infty$ is dense and invariant under every $\Pi(\exp tX)$. The dense invariant-domain criterion of Theorem 15.11(b), applied with $D = \mathcal{H}^\infty$, shows that the closure of $d\Pi(X)$ is B_X. The uniqueness statement of Theorem 15.11(c) then gives $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$. ∎

In the physical convention above, the *physical generator* of $t \mapsto \Pi(\exp tX)$ is $A_X = iB_X$, so $\Pi(\exp tX) = e^{-itA_X}$. Since B_X is skew-adjoint, A_X is self-adjoint. For translations and $X = 1 \in \mathbb{R}$, we have $B_1 = A$. With $\hbar = 1$, the operator $A_1 = iA$ is the momentum operator P, formally $-i\,d/dx$.

15.5 Translations on $L^2(\mathbb{R})$

The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the space of smooth $f\colon \mathbb{R}\to\mathbb{C}$ such that

$$\sup_{x\in\mathbb{R}} |x^k f^{(j)}(x)| < \infty \qquad \text{for all } j,k\ge 0.$$

It contains $C_c^\infty(\mathbb{R})$ and $e^{-x^2/2}$. Let A be the generator of λ.

Lemma 15.13. *Let $f\in C^2(\mathbb{R})$ with f and f' square integrable. Suppose $|f''(y)|\le C(1+y^2)^{-1}$ for all y. Then $f\in D(A)$ and $Af=-f'$.*

Proof. Fix $0<|h|\le 1$. Taylor's formula with integral remainder, for $s\mapsto f(x-sh)$ on $[0,1]$, gives

$$f(x-h)=f(x)-hf'(x)+h^2\int_0^1 (1-s)f''(x-sh)\,ds.$$

So the error function is

$$q_h(x)=\frac{f(x-h)-f(x)}{h}+f'(x)=h\int_0^1(1-s)f''(x-sh)\,ds,$$

and $|q_h(x)|\le \frac{|h|}{2}\sup_{|y-x|\le 1}|f''(y)|$. If $|y-x|\le 1$, then $x^2\le (|y|+1)^2\le 2y^2+2$, so $1+x^2\le 3(1+y^2)$. Therefore

$$|q_h(x)|\le \frac{3C|h|}{2(1+x^2)}, \qquad \|q_h\|^2\le \frac{9C^2h^2}{4}\int_{\mathbb{R}}\frac{dx}{(1+x^2)^2}=\frac{9\pi C^2h^2}{8}.$$

Thus $\left\|\frac{\lambda(h)f-f}{h}-(-f')\right\|=\|q_h\|\to 0$. ∎

The essential hypothesis is the weighted bound $|f''(y)|\le C(1+y^2)^{-1}$. The difference quotient in the proof converges pointwise for every C^1 function f. Pointwise convergence alone does not give convergence in norm. The weighted bound makes the error q_h at most $|h|$ times a fixed square-integrable function, so the norm of q_h is $O(|h|)$.

Proposition 15.14. *The space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, invariant under every $\lambda(t)$, and contained in $L^2(\mathbb{R})^\infty$. For $f\in\mathcal{S}(\mathbb{R})$ we have $d\lambda(1)f=-f'$. The closure of $B=-d/dx$ with domain $\mathcal{S}(\mathbb{R})$ is the skew-adjoint generator A.*

Proof. Invariance. Put $y=x-t$. Expanding $x^k=(y+t)^k$ bounds $|x^kf^{(j)}(x-t)|$ by a finite sum of terms $\binom{k}{i}|t|^{k-i}|y^if^{(j)}(y)|$.

Smoothness. Each $f\in\mathcal{S}(\mathbb{R})$ satisfies the hypotheses of the Taylor-remainder lemma, Lemma 15.13. Indeed $(1+y^2)f$, $(1+y^2)f'$ and $(1+y^2)f''$ are bounded. So f and f' are square integrable, and $|f''|\le C(1+y^2)^{-1}$. In particular $\mathcal{S}(\mathbb{R})\subset L^2(\mathbb{R})$. By the lemma, $t\mapsto\lambda(t)f$ is differentiable at 0 with derivative $-f'$. At any t_0,

$$\frac{\lambda(t_0+h)f-\lambda(t_0)f}{h}=\lambda(t_0)\,\frac{\lambda(h)f-f}{h}\longrightarrow\lambda(t_0)(-f'),$$

because $\lambda(t_0)$ is bounded. The derivative $t\mapsto\lambda(t)(-f')$ is continuous by the strong continuity of translations, Proposition 15.3. Since $-f'\in\mathcal{S}(\mathbb{R})$, we may repeat. The k-th derivative is $t\mapsto\lambda(t)\big((-1)^kf^{(k)}\big)$, so f is a smooth vector.

Density. Fix $\varphi \in C_c^\infty(\mathbb{R})$ with $\varphi \geq 0$, support in $[-1,1]$ and $\int \varphi = 1$. Put $\varphi_n(t) = n\varphi(nt)$. For $f \in C_c(\mathbb{R})$ vanishing outside $[-R,R]$, let

$$f_n(x) = \int_{\mathbb{R}} \varphi_n(x-y) f(y)\, dy = \int_{\mathbb{R}} \varphi_n(t) f(x-t)\, dt.$$

The first form shows $f_n \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$, with support in $[-R-1, R+1]$. The second form gives

$$|f_n(x) - f(x)| \leq \sup_{|t| \leq 1/n} |f(x-t) - f(x)|,$$

which tends to 0 uniformly in x because f is uniformly continuous. Since f_n and f vanish outside $[-R-1, R+1]$, it follows that $f_n \to f$ in $L^2(\mathbb{R})$. Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, so is $\mathcal{S}(\mathbb{R})$.

Closure. By the three steps above, $D = \mathcal{S}(\mathbb{R})$ is a dense invariant subspace of $D(A)$. So the dense invariant-domain criterion of Theorem 15.11(b) shows that the closure of B is A. ∎

The function f_n is the Gårding vector $\lambda(\varphi_n)f$ written pointwise.

Example 15.15. The vectors w and h, and the operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$, show where the domains lie.

(i) Let $w(x) = (1+x^2)^{-1}$. Then

$$w'(x) = \frac{-2x}{(1+x^2)^2}, \qquad w''(x) = \frac{6x^2 - 2}{(1+x^2)^3}.$$

Since $|6x^2 - 2| \leq 6(1+x^2)$, we get $|w''(x)| \leq 6(1+x^2)^{-2} \leq 6(1+x^2)^{-1}$. Both w and w' are square integrable. By the Taylor-remainder lemma, Lemma 15.13, $w \in D(A)$ and

$$(Aw)(x) = \frac{2x}{(1+x^2)^2}.$$

But $x^2 w(x) \to 1$, so $x^4 w(x)$ is unbounded and $w \notin \mathcal{S}(\mathbb{R})$.

(ii) The operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$ is the restriction of $d\lambda(1)$ to $\mathcal{S}(\mathbb{R})$ by Proposition 15.14. So B is skew-symmetric by the skew-symmetry of $d\lambda(1)$ on smooth vectors, Proposition 15.5(d). It is not skew-adjoint. Indeed $B \subset A$, the domain $\mathcal{S}(\mathbb{R})$ is dense, and $A^* = -A$. So $-A = A^* \subset B^*$. Hence $w \in D(A) \subset D(B^*)$, while $w \notin D(B)$. So $D(B^*) \neq D(B)$, and therefore $B^* \neq -B$.

(iii) Let $h = 1_{[0,1]}$. For $0 < t \leq 1$, up to finitely many points,

$$\lambda(t)h - h = 1_{[t,1+t]} - 1_{[0,1]} = 1_{(1,1+t]} - 1_{[0,t)}, \qquad \|\lambda(t)h - h\|^2 = 2t.$$

For $-1 \leq t < 0$ the same computation gives $2|t|$. Therefore

$$\left\| \frac{\lambda(t)h - h}{t} \right\| = \sqrt{\frac{2}{|t|}} \longrightarrow \infty.$$

Convergent quotients would stay bounded near 0. So $h \notin D(A)$. Smooth vectors lie in $D(A)$ by Corollary 15.12, so also $h \notin L^2(\mathbb{R})^\infty$.

Together these give

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})^\infty \subset D(A), \qquad w \in D(A) \setminus \mathcal{S}(\mathbb{R}), \qquad h \in L^2(\mathbb{R}) \setminus D(A).$$

Exercise 15.2 gives a tent function in $D(A)$ that is not a smooth vector. The momentum operator $P = iA$ has domain $D(A)$, not $\mathcal{S}(\mathbb{R})$.

15.6 Self-checks

1. Let $\dim \mathcal{H} < \infty$. What are $\mathcal{H}^\infty$ and $D(B_X)$?
   Answer. Both equal $\mathcal{H}$. The homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ is continuous, hence smooth (Lecture 3), so every vector is smooth. Since $d\Pi(X)$ is then defined on all of $\mathcal{H}$, its closure B_X also has domain $\mathcal{H}$.
2. Let $f(x) = xe^{-x^2/2}$. Compute Af and $\langle f, Af\rangle$.
   Answer. Since $f \in \mathcal{S}(\mathbb{R})$, the Schwartz-space proposition, Proposition 15.14, gives $Af = -f' = (x^2 - 1)e^{-x^2/2}$. Then $\langle f, Af\rangle = \int_{\mathbb{R}} x(x^2 - 1)e^{-x^2}\,dx = 0$, because the integrand is odd. This is consistent with skew-symmetry, which forces $\langle f, Af\rangle$ to be purely imaginary.
3. Let D be the space of smooth functions with compact support in $(0, \infty)$. Then $D \subset D(A)$. Does the dense invariant-domain criterion of Theorem 15.11(b) apply to D?
   Answer. No. The space D is not dense, since every element is orthogonal to $1_{[-1,0]} \neq 0$. It is not invariant either. Given $t < 0$, choose a nonzero $f \in D$ supported in $(0, |t|)$. Then $\lambda(t)f$ is supported in $(t, 0)$, so $\lambda(t)f \notin D$.

15.7 Summary

Unitary representations are required to be strongly continuous, not norm continuous. Smooth vectors form a dense invariant subspace $\mathcal{H}^\infty$. On it, $d\Pi$ is a representation of $\mathfrak{g}$ by skew-symmetric operators. Stone's theorem attaches a skew-adjoint generator to each strongly continuous unitary representation of $\mathbb{R}$. The closure of $d\Pi(X)$ on $\mathcal{H}^\infty$ is this generator. For translations, $-d/dx$ on $\mathcal{S}(\mathbb{R})$ is skew-symmetric but not skew-adjoint. Its closure has the strictly larger domain $D(A)$, and $D(A)$ is still not all of $L^2(\mathbb{R})$.

15.8 Reading

- Hall, *Quantum Theory for Mathematicians* (2013): unbounded self-adjoint operators and Stone's theorem, Section 10.2.
- Reed–Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Section VIII.4, especially Theorems VIII.10–11 on dense invariant domains.
- Knapp, *Representation Theory of Semisimple Groups* (2001), Sections III.3–4: smooth vectors, Gårding vectors and the action of $\mathfrak{g}$.
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